

NeuroCBT: Neuroadaptive Large Language Model for EEG-Guided Cognitive Behavioral Therapy

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Abstract—We present NeuroCBT, a neuroadaptive conversational AI framework that integrates consumer-grade EEG signals with a multi-agent CBT architecture grounded in the Cognitive Triangle. The system consists of dedicated agents for thoughts, emotions, and behaviors, alongside an EEG Analysis Agent that adapts responses to real-time physiological states. To enable scalable deployment, we also develop NeuroCBT-LLM, a lightweight end-to-end model trained on a clinically reviewed 6,000-sample EEG-text dataset. Experiments across GPT-4o, Qwen 2.5-7B, LLaMA 3-8B, and Mistral-7B show that our framework consistently improves empathy, guidance, and interactivity. Notably, Qwen 2.5-7B with EEG achieved 7× higher Elo-per-second efficiency than GPT-4o, and ablation studies confirmed the distinct roles of each module. A randomized EEG control validated the system’s physiological sensitivity, while a human study ($N = 32$) showed significant gains in perceived empathy and emotional congruence. These findings demonstrate NeuroCBT’s potential as a scalable, affect-aware framework for real-time mental health interventions.

Index Terms—EEG, CBT, LLM, Multi-Agent, Neuroadaptive AI

I. INTRODUCTION

Cognitive Behavioral Therapy (CBT) [1], [2] is a representative psychological treatment technique with proven scientific effectiveness for various mental health issues, showing long-term therapeutic effects for various symptoms such as depression, anxiety disorders, and post-traumatic stress disorder (PTSD) [3]. CBT induces emotional and behavioral changes by examining the patient’s thought patterns and restructuring irrational thoughts into realistic ones [4]. This structured approach is highly suitable for digitization and automation due to its quantitative evaluation and repeatable application, and it is currently being expanded to further enhance treatment effects by combining it with physiological signal-based technologies. In particular, neurofeedback has demonstrated the potential to enhance therapeutic effects by measuring and visualizing biological signals such as electroencephalography (EEG) in real time, helping users recognize and regulate their own states through brain function regulation.

EEG-based neurofeedback CBT has been reported to have a positive effect on improving patients’ attention, emotional stability, and self-regulation abilities, and some studies have shown faster and more sustained treatment effects compared to conventional CBT [5], [6]. However, this approach requires expensive EEG equipment, controlled measurement environments, and constant intervention by specialized personnel, making it available primarily in offline clinical settings [7]. These physical constraints severely limit the scalability of neurofeedback CBT in an era where demand for telepsy-

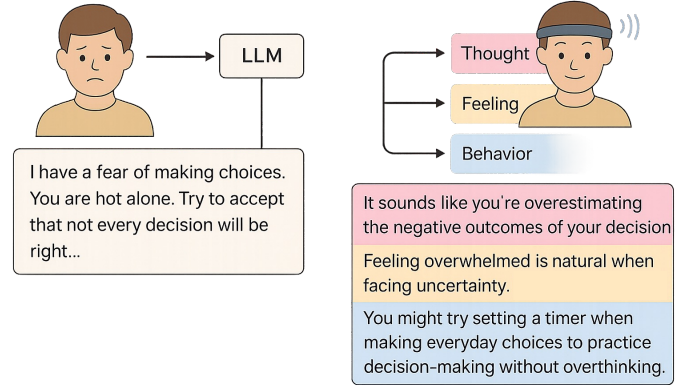


Fig. 1. Comparison between traditional single-agent CBT chatbot (left) and proposed NeuroCBT framework (right).

chotherapy and digital therapeutics is on the rise. Ultimately, despite the clinical effectiveness of CBT and the substantial contributions of neurofeedback, the combination of the two remains limited to a small number of clinical settings, necessitating a technological transition for online application.

Against this backdrop, recent advances in AI technology are enabling the online implementation of CBT. In particular, large language models (LLMs) can maintain the flow of structured conversations and automatically guide patients through core CBT techniques such as cognitive restructuring, emotional expression, and behavioral planning [8], [9], [10]. In fact, with the commercialization of CBT chatbots such as Woebot and Wysa, the possibility of mental health interventions that are not constrained by time or location is rapidly becoming a reality [11], [12]. However, these systems mostly rely on text input and have structural limitations in that they cannot reflect nonverbal and physiological cues such as the user’s psychological state, cognitive load, and emotional responses. This leads to various problems, such as inaccurate empathetic responses, failure to respond to crisis situations, and mismatched treatment timing, which hinder the precision and reliability required in actual clinical practice [13]. The structural limitations of traditional chatbot methods can be seen in the left side of Figure 1, which visually represents the process of generating uniform responses based solely on text input.

To overcome these limitations, this study proposes NeuroCBT, a neuroadaptive CBT framework that integrates EEG-

based real-time physiological signal analysis and a multi-LLM agent structure. Specifically, this framework designs three language model agents, each responsible for thoughts, feelings, and behaviors, respectively, in accordance with the Cognitive Triangle, the core theoretical foundation of CBT, as shown on the right side of Figure 1. Each agent communicates with others to establish intervention strategies tailored to their respective domains, and the results are integrated with user states extracted in real-time from EEG by the EEG Analysis Agent, providing the most appropriate CBT feedback to the user. This structure precisely reflects the theoretical principles of CBT in the actual response generation mechanism and integrates real-time EEG-based neurofeedback information into the treatment flow, thereby realizing precise and adaptive interventions utilizing the language processing capabilities of LLM.

Additionally, to realize the proposed framework as an expandable and economically deployable end-to-end model, we developed NeuroCBT-LLM, which was trained using EEG signals and CBT response pairs. This model is designed to simultaneously perform EEG interpretation and CBT response generation within a single LLM without a multi-agent structure, offering the advantage of easy deployment in mobile environments or cloud-based services. This enables neurophysiology-based CBT intervention without the need for a high-cost multi-module system, thereby significantly improving the accessibility and practicality of digital mental health interventions.

The main contributions of this paper are as follows:

- We propose NeuroCBT, the first online CBT framework that integrates a multi-agent structure based on the Cognitive Triangle theory and EEG-based neurofeedback information.
- By developing an end-to-end NeuroCBT-LLM that integrates EEG interpretation and CBT response generation within a single language model, we have simultaneously ensured deployment efficiency and cost-effectiveness.
- Through various user scenarios and control experiments, we demonstrated that the proposed NeuroCBT framework and NeuroCBT-LLM outperform existing text-based CBT chatbots in terms of emotional alignment, empathetic responses, and user satisfaction.

II. RELATED WORK

A. Neurofeedback-based CBT

Neurofeedback, which visualizes EEG signals in real time to help individuals regulate brain states, has shown lasting benefits for ADHD, anxiety, and depression [7], [14], and studies indicate that training prefrontal θ/β ratio can enhance CBT techniques like cognitive restructuring [15]. However, traditional neurofeedback relies on costly multi-channel EEG systems, expert supervision, and in-person sessions over several weeks, limiting scalability [16], [17], [18]. Attempts at home-based training with low-channel devices have faced issues like electrode misplacement and noise, yielding limited clinical

success [19]. Thus, despite proven efficacy, neurofeedback-CBT requires a digital transformation for remote, real-time delivery that integrates physiological signals into therapeutic interventions.

B. Chatbot and LLM-based CBT

Text-based CBT chatbots have improved accessibility and scalability, with Woebot reducing depression scores in a two-week RCT [20], and platforms like Wysa [21] and Youper [22] serving millions by automating mindfulness and behavioral tasks. Recently, LLM-based chatbots (e.g., GPT-4) have shown strong conversational and contextual abilities in clinical evaluations [23]. However, these systems rely solely on text, failing to capture users' cognitive and physiological states, leading to safety concerns such as missed suicidal cues and hallucinated responses [24], [25]. Excluding nonverbal signals like EEG and HRV limits empathy timing and feedback precision, highlighting the need for adaptive, closed-loop LLM interventions integrating real-time neural indicators.

C. Multi-agent-based consultation system

An approach using role-based multi-agent dialogue to handle complex tasks is emerging. Frameworks like Auto-GPT and ChatDev model virtual software companies where specialized LLM agents collaborate to complete tasks previously difficult for a single model [26]. In mental health, combining emotion analysis agents with CBT dialogue agents has improved counseling quality [27], and multi-agent systems supporting counselors—where one agent summarizes conversations and another suggests responses—have shown human-level empathy and efficiency gains [28].

However, prior studies mainly share linguistic information or a single emotional score among agents [29], [30], lacking integration of rapidly changing physiological signals like EEG/HRV or explicit modeling of the CBT cognitive triangle (thoughts, feelings, behaviors). To enable multi-modal, neurofeedback-based closed-loop CBT, a new design is required. NeuroCBT addresses this gap by sharing EEG-based user states in real time with thought, feeling, and behavior agents, while the EEG Analysis Agent integrates these signals to generate optimal therapeutic dialogue—providing the first framework that unifies effectiveness, precision, and scalability.

III. NEUROCBT FRAMEWORK

A. Overall Architecture

NeuroCBT processes text and EEG inputs in parallel through a multi-agent approach. EEG signals, collected via a 4-channel Muse-S headband [31] during natural speech, are synchronized with text within 5ms using a shared timestamp. EEG data undergo real-time preprocessing into low-dimensional vectors, while utterances go to the Cognitive-Triangle layer, which includes three LLM-based agents for thought, feeling, and behavior with role-specific prompts. The EEG Agent integrates physiological indicators with agent discussions to generate an adjustment report, which, along

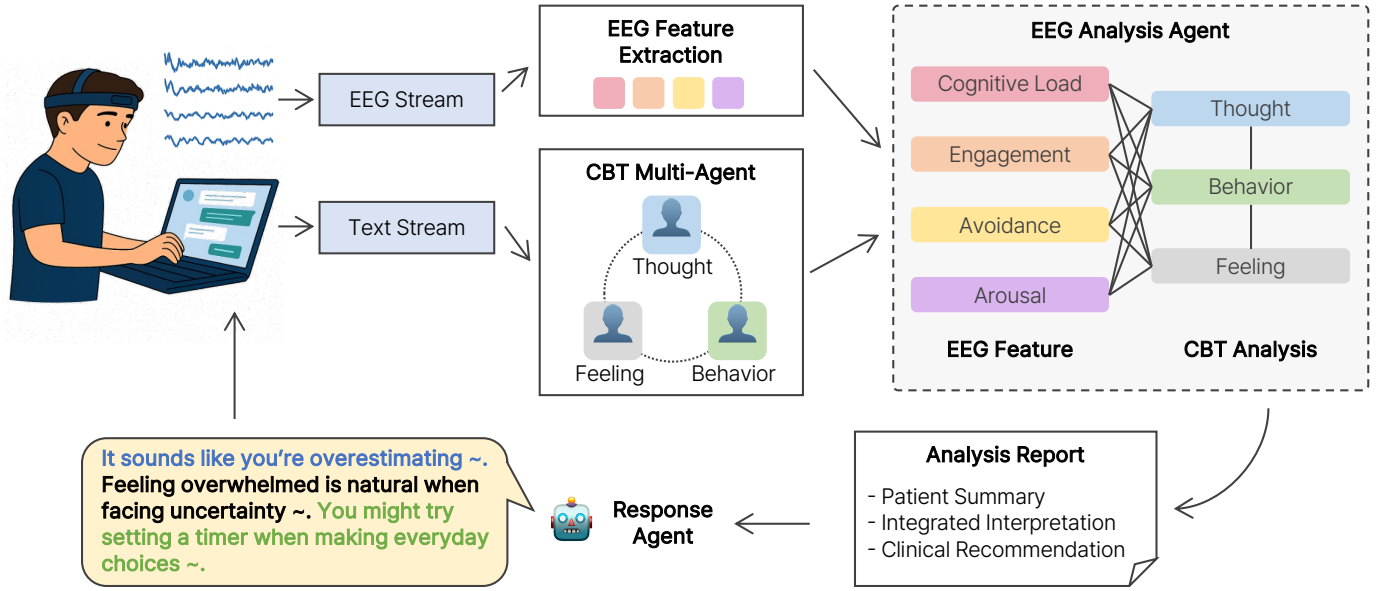


Fig. 2. NeuroCBT has a closed-loop structure in which text and EEG streams are processed in parallel, multiple CBT agents (Thought-Feeling-Behavior) are analyzed, real-time EEG indicators (Cognitive Load, Engagement, Avoidance, Arousal) are integrated in the EEG Analysis Agent, and based on this, the Response Agent returns personalized treatment responses.

with the user's input, forms the final LLM prompt for therapeutic response—combining multi-agent reasoning with LLM fluency.

B. EEG Feature Extraction Layer

Real-time EEG was recorded at 256Hz using a 4-channel Muse-S headband (AF7-AF8, TP9-TP10) and streamed via Brainflow. Preprocessing and feature extraction were implemented in Python. Muse-S uses dry electrodes for easy wear and cleaning, and its prefrontal channels (AF7-AF8) provide stable signals sufficient for attention and emotion estimation needed in CBT. Preprocessing involved a 0.5–45Hz bandpass filter, a 50Hz notch filter, Artifact Subspace Reconstruction (ASR) for motion/blink noise removal, and common average reference (CAR) for spatial noise reduction. Signals were segmented with a 2s sliding window (0.4s overlap), Z-normalized, and scaled to 0–1. Four EEG indicators were extracted using Welch's method, where $\sigma(\cdot)$ is the sigmoid function and $P_x(c)$ denotes relative band power:

$$\text{Cognitive Load} = \frac{P_\theta(\text{AF7}) + P_\theta(\text{AF8})}{2}, \quad (1)$$

$$\text{Engagement} = \frac{P_\beta}{P_\alpha + P_\theta}, \quad (2)$$

$$\text{Avoidance} = \sigma(-[P_\alpha(\text{AF7}) - P_\alpha(\text{AF8})]), \quad (3)$$

$$\text{Arousal} = \frac{P_\beta}{P_\alpha}. \quad (4)$$

These features capture cognitive load, engagement, avoidance tendency, and arousal, serving as inputs for the EEG Analysis Agent. All metrics rely on prefrontal frequency characteristics, ensuring compatibility with low-channel devices

like Muse-S and enabling real-time, neurofeedback-driven CBT in online environments.

C. Cognitive-Triangle Agent Layer

This layer was designed to implement Beck's Cognitive Triangle—a model in which thought, feeling, and behavior are interconnected with the situation in a cyclical manner—as a dialogue mechanism. Furthermore, the dialogue stages of Explorations, Emotional Reactions, and Interpretations identified by Sharma et al. through their analysis of medical consultation corpora are structurally consistent with the sequence and structure of exploring the three axes of thought, emotion, and behavior. [27], [32] Accordingly, the same language model was divided into three instances, and each instance was assigned a system prompt corresponding to the role of thought, emotion, or behavior. The input for each agent consists solely of the client's utterances and prompts, and EEG information is not incorporated at this stage.

The first Thought Agent corresponds to the Explorations stage. The agent reconstructs the situation and auto-thoughts from the client's utterances and checks for distortions based on the 10 cognitive distortions proposed by Beck. If a distortion is suspected, the agent applies Socratic questioning to present an alternative perspective in 30 words or less. The purpose of the Thought Agent is to clarify "what happened and what thoughts arose" at the hypothesis level, laying the foundation for subsequent emotional and behavioral inferences.

The second Feeling Agent corresponds to the Emotional Reactions stage. This agent estimates emotions and their intensity based on Plutchik's 8-category system [33] and Gross's emotion component theory [34]. The core function of the Feeling Agent is to accurately name the emotions revealed in the client's utterances and to establish emotional safety.

The third Behavior Agent corresponds to the Interpretations stage and estimates behavioral patterns that may derive from emotional responses based on the behavioral experimental model [35], [36]. The agent describes behavioral hypotheses such as “She tends to postpone decisions to avoid negative outcomes,” and if avoidance tendencies are strongly evident, it automatically lowers the difficulty level to easy to induce initial success experiences. The Behavior Agent prepares specific behavioral strategies that the client can apply in real life.

The messages from the three agents are transmitted to the next stage, the EEG Analysis Agent. This step-by-step exploration structure reduces the context confusion and hallucination possibilities often seen in single language models and enables early detection of inconsistencies between thoughts, emotions, and behaviors. At the same time, the rich dialogue records generated by the agents provide sufficient evidence for deriving customized rules when combined with subsequent EEG data, thereby strengthening the theoretical consistency and clinical usefulness of the NeuroCBT pipeline.

D. EEG Analysis Agent

The EEG Agent is a module that combines linguistic reasoning and neurophysiological information in the NeuroCBT pipeline. This module accepts four EEG indicators (Cognitive Load, Engagement, Avoidance, and Arousal) generated by the EEG Feature Extraction Layer and the entire conversation record exchanged between Cognitive-Triangle Agents as simultaneous inputs. EEG signals are transmitted with each dialogue record within a 5 ms range from the time of the dialogue, reflecting the state at that time. By combining the two types of information on the same timestamp axis, the EEG Analysis Agent can detect discrepancies between the psychologically inferred state and the neurophysiologically observed state, and integrate them into a refined report.

The EEG Analysis Agent is designed based on the prognostic value of cognitive load, engagement, avoidance, and arousal, which have been repeatedly confirmed in CBT research. High cognitive load leads to fixation on thought distortions [37], [38], low Engagement suggests emotional avoidance or distrust of counseling [39], and the Avoidance value calculated from left-right prefrontal α asymmetry predicts negative emotion bias and behavioral avoidance levels [40]. These indicators provide momentary fluctuations that are difficult to capture with text alone, serving as objective evidence to validate and supplement the reliability of hypotheses formed by Cognitive-Triangle Agents. In other words, the EEG Analysis Agent functions as a quantitative module that complements the blind spots of language interpretation.

The output is returned in the form of a structured report rather than free narrative. The report presents (i) the latest EEG values and client utterances, (ii) physiological-language integrated interpretation, and (iii) clinical recommendations item by item. With the EEG Analysis Agent, NeuroCBT expands beyond a simple text chatbot to become a real-time physiological-language integrated closed-loop treatment system. This has significant clinical implications in that (1)

thought, emotion, and behavior hypotheses are cross-validated with physiological indicators to reduce interpretation errors, (2) customized behavioral tasks and relaxation techniques are automatically adjusted to the user’s actual arousal and avoidance levels, and (3) data on changes between sessions can be accumulated and used for long-term prognosis prediction.

IV. NEUROCBT LLM

A. NeuroCBT Datasets

To address the lack of multimodal data combining language and physiology, we built the NeuroCBT dataset using 200 high-difficulty CBT-BENCH Level 3 scenarios adapted to the domestic context [41]. Each scenario generated triples (q_i, e_i, v_i) , where q_i = client utterance, e_i = EEG indicators Cognitive Load, Engagement, Avoidance, Arousal, and v_i = NeuroCBT response, producing 1,200 initial turns.

We expanded diversity through LLM-based dialogue augmentation [42]. (i) Emotion-style transformation adjusted EEG arousal/emotion values for the same text (e.g., “I’m worried about the exam” → high-arousal anxiety vs. low-arousal depression) and regenerated responses. EEG values were also added to neutral statements to model implicit emotions. (ii) Meaning-preserving paraphrasing diversified vocabulary and syntax while maintaining CBT style using GPT-3.5 (e.g., “I’m afraid of failing” → “I’m worried about messing up”). After two-step augmentation, 6,000 triples were obtained, each with NeuroCBT-regenerated responses for logical consistency. CBT and neurofeedback experts validated the dataset for clinical reliability.

B. Fine-tuning Strategy

We fine-tuned Qwen 2.5-7B, LLaMA 3-8B, and Mistral-7B-v0.3, creating three variants: (i) Base (pre-trained), (ii) Text-only (trained on text pairs), and (iii) NeuroCBT (EEG-augmented). In (iii), EEG info was wrapped in special tokens before user text, with the instruction “Plan first; then respond” to jointly consider physiological and CBT reasoning. The learning method combines instruction fine-tuning with LoRA (Low-Rank Adaptation) [43]. LoRA adds low-rank matrices while freezing base weights for parameter efficiency, and instruction learning aligns outputs to CBT-EEG objectives.

V. EXPERIMENT

A. Baseline Configuration

The experiment was structured along two axes.

First, to evaluate the effectiveness of the **NeuroCBT framework** itself, we compared two conditions—*Base (without framework)* and *Framework Applied (Multi-agent + EEG)*—across four models: GPT-4o, Qwen 2.5-7B, LLaMA 3-8B, and Mistral-7B-v0.3. For Qwen 2.5-7B, which showed the most prominent performance difference, we additionally compared the *single-agent* and *multi-agent* configurations to quantify the contribution of role division.

Second, to verify the learning effect of **NeuroCBT-LLM**, we fine-tuned three models (Qwen 2.5-7B, LLaMA 3-8B, Mistral-7B-v0.3) under two input conditions: *Text-only* and

TABLE I
NEUROCBT FRAMEWORK OVERALL RESULT

Model	Framework	Analytical (Z-Score)	Empathy (Z-Score)	Guidance (Z-Score)	Interactivity (Z-Score)
GPT-4o	Baseline	1483 (0.12)	1456 (-0.23)	1471 (0.08)	1489 (0.18)
	NeuroCBT	1641 (1.89)	1645 (1.95)	1628 (1.78)	1635 (1.85)
Mistral	Baseline	1445 (-0.31)	1419 (-0.65)	1432 (-0.42)	1451 (-0.58)
	NeuroCBT	1550 (0.87)	1568 (0.93)	1545 (0.82)	1558 (0.89)
LLaMA 3	Baseline	1438 (-0.38)	1412 (-0.72)	1425 (-0.49)	1444 (-0.65)
	NeuroCBT	1520 (0.57)	1535 (0.61)	1518 (0.55)	1527 (0.58)
Ours(Qwen 2.5)	Baseline	1461 (-0.08)	1434 (-0.42)	1448 (-0.15)	1467 (-0.29)
	NeuroCBT	1574 (1.12)	1589 (1.18)	1571 (1.08)	1582 (1.15)

With EEG. Fine-tuning was conducted using the **NeuroCBT dataset** introduced in Section 4.1, and we maintained identical hyperparameters to ensure comparability across models.

B. Evaluation Metrics

For quantitative evaluation, we adopted the **ELO Rating** method [44]. Using Claude 3.5 Sonnet as the evaluator, two responses generated for the same prompt were compared across four dimensions—*Analytical*, *Empathy*, *Guidance*, and *Comprehensive*—to determine the superior response for each. The Elo score for each dimension started at an initial value of 1500 and was updated based on a win-loss formula; in cases where the response quality was similar, a *tie* was recorded. Each response pair was evaluated five times, and the average was used. To facilitate final comparisons, the Elo scores for each dimension were normalized using the Z-score method.

For qualitative evaluation, we conducted a user study with 32 participants. After completing a scenario-based CBT conversation, participants rated six items on a 7-point Likert scale: empathy, emotional alignment, helpfulness of advice, emotional stabilization, system trustworthiness, and willingness to reuse the system.

C. Small-Scale Expert Evaluation

A small-scale expert evaluation was conducted with five clinical professionals: one doctoral-level neurofeedback-CBT specialist and four master’s-level practitioners, all experienced with digital mental health tools. Using a randomized, single-blind design, each expert evaluated 20 dialogue pairs (100 unique scenarios), each consisting of a NeuroCBT-generated and a human therapist response presented in random order, with evaluators blinded to the source. Sessions followed a standardized 90-minute protocol, including a 15-minute briefing on evaluation criteria and CBT standards, a 60-minute independent rating phase, and a 15-minute semi-structured interview for qualitative feedback. Responses were assessed on a 5-point Likert scale across four dimensions, with clearly defined behavioral anchors.

D. Implementation and Training Environment

All experiments were implemented using PyTorch, and computations were performed on two NVIDIA H100-80GB GPUs. Fine-tuning was carried out using instruction fine-tuning and LoRA adapters (rank = 16), with optimization performed by minimizing the cross-entropy loss. The shared hyperparameters were a learning rate of 2×10^{-5} , batch size of 16, and training over 10–20 epochs. This unified configuration allowed performance differences across models and conditions to be interpreted independently of network architecture or other dataset-related factors.

1) Ablation Study:

VI. RESULTS

A. Performance of the NeuroCBT Framework

1) *Comparison of NeuroCBT Framework Performance:* Table I clearly demonstrates that the NeuroCBT framework consistently yields positive effects across four different language models. For GPT-4o, the baseline condition resulted in an average score of approximately 1475 across the four evaluation metrics (roughly within the Z-score range of 0 ± 0.3). However, after applying the framework, the average score increased to 1637 (Analytical: 1641, Empathy: 1645, Guidance: 1628, Interactivity: 1635), showing a substantial and meaningful improvement of about 158 points (1.95 standard deviations) per metric.

Among the open-source models, Qwen 2.5-7B stood out. Its Empathy score improved from 1434 to 1589, and its Analytical score increased from 1461 to 1574, achieving an overall average of 1577 across the four metrics—a relative improvement of approximately 7.7% over the baseline. While Mistral-7B-v0.3 and LLaMA 3-8B also showed improvements of 105 and 82 points on average, respectively, their absolute performance did not reach the same level as Qwen 2.5-7B.

These results suggest that the NeuroCBT framework not only enhances performance reliably across both proprietary large-scale models and lightweight open-source models, but

TABLE II
ABLATION STUDY OF CBT AGENT AND EEG AGENT (BASED ON GPT-4o)

System Configuration	Analytical	Empathy	Guidance	Interactivity
Baseline (No Agents)	1483	1456	1471	1489
CBT Agent Only	1558	1521	1578	1542
EEG Agent Only	1524	1501	1512	1530
Full System (Both)	1641	1645	1628	1635
<i>CBT Agent Contribution</i>	+75	+65	+107	+53
<i>EEG Agent Contribution</i>	+41	+45	+41	+41
<i>Synergy Effect</i>	+42	+79	+9	+53

also that Qwen 2.5-7B is particularly effective at leveraging the combined EEG-text input modality.

The ablation study presented in Table II disentangles the individual contributions and interactions of the two core modules. When only the CBT Agent was activated, GPT-4o’s Guidance score increased from 1471 to 1578—a gain of 107 points—whereas the improvement in Empathy was limited to just 65 points. In contrast, using only the EEG Agent resulted in more balanced improvements across all four dimensions, each rising by approximately 45 points. Notably, the Empathy score increased by 45 points, reaching about 70% of the improvement achieved by the CBT Agent alone.

In the full configuration where both modules were used, additional gains of 79 and 53 points were observed in the Empathy and Interactivity dimensions, respectively. This suggests a multiplicative synergy between the two modules in enhancing emotional understanding and conversational naturalness. The logical flow enabled by structured cognitive exploration complements the emotional context provided by real-time physiological signals, compensating for each other’s weaknesses. As a result, the overall average Elo score increased by 162 points over the baseline, which exceeds the simple sum of their individual contributions (+75 and +41) by 45 points—strongly supporting the design rationale of the integrated system.

B. Evaluation of NeuroCBT-LLM Performance

Table III illustrates the effectiveness of fine-tuning with the NeuroCBT dataset. Qwen 2.5-7B showed a notable improvement, with its average score rising from a baseline of 1461 to 1548, a gain of 87 points. Compared to the Text-only version, it also achieved an additional 64-point improvement. The most significant enhancement was observed in the Empathy dimension, where the score increased from 1434 to 1563 (+129), confirming that EEG-based emotional context learning directly contributes to higher-quality empathetic responses. Mistral-7B-v0.3 also improved by an average of 83 points, and LLaMA 3-8B by 57 points. However, their absolute post-finetuning scores remained lower than Qwen 2.5-7B, reaching 1528 and 1495 respectively.

Efficiency analysis considering both response latency and performance (Table IV) demonstrates the real-time applicabil-

ity of the NeuroCBT-LLM approach. While GPT-4o exhibits an increased response time from 8.2 seconds (baseline) to 15.2 seconds when NeuroCBT is applied, Qwen 2.5-NeuroCBT-LLM maintains a high performance of approximately 1579 points within just 2.1 ± 0.3 seconds. This yields an Elo per second efficiency of 751.9, which is nearly 7 times higher than that of GPT-4o-NeuroCBT. Since response times under 2 seconds are widely regarded as the threshold for maintaining user engagement in real-time counseling, these results indicate that Qwen 2.5-NeuroCBT-LLM achieves a practical balance of speed and quality suitable for deployment in clinical environments.

The random EEG control experiment in Table V demonstrates that the model’s interpretation of physiological signals is not due to statistical chance. When the EEG vectors were replaced with random values, the average Elo score dropped from 1579 to 1438—a decrease of 142 points. The Empathy score, in particular, declined from 1589 to 1404, a reduction of 185 points. Given that the text input remained identical while only the signal values were altered, the substantial drop in performance clearly indicates that the model genuinely utilizes EEG indicators as critical cues for emotional interpretation.

C. Human Evaluation

A 7-point Likert-scale evaluation with 32 participants showed that NeuroCBT substantially improved the user experience compared to a GPT-based chatbot. Specifically, *Empathy* and *Emotional Alignment* increased by about 33% (+1.54 points), while *Support for Emotional Stabilization* rose by +1.40 points. Although improvements in *Advice Usefulness* and *Trustworthiness* were smaller (below 1 point), they were still statistically significant, indicating GPT-based systems already perform reasonably well in generating advice.

Qualitative feedback reinforced these results: 84% of participants preferred the EEG-integrated system, noting comments such as “it actually understands me” and “it gives suggestions that match my state.” These findings suggest that the real-time emotional reflection provided by the EEG Agent directly enhances users’ sense of being understood and supported.

TABLE III
NEUROCBT LLM COMPARISON

Model	Dataset	Analytical (Z-Score)	Empathy (Z-Score)	Guidance (Z-Score)	Interactivity (Z-Score)
Mistral	Baseline	1445 (-0.45)	1419 (-0.62)	1432 (-0.48)	1451 (-0.41)
	Text-only	1459 (-0.31)	1442 (-0.39)	1455 (-0.25)	1461 (-0.31)
	NeuroCBT	1528 (0.76)	1545 (0.84)	1522 (0.73)	1535 (0.79)
LLaMA 3	Baseline	1438 (-0.52)	1412 (-0.69)	1425 (-0.55)	1444 (-0.48)
	Text-only	1440 (-0.50)	1425 (-0.56)	1438 (-0.42)	1447 (-0.45)
	NeuroCBT	1495 (0.43)	1510 (0.49)	1493 (0.40)	1502 (0.46)
Ours(Qwen 2.5)	Baseline	1461 (-0.29)	1434 (-0.52)	1448 (-0.35)	1467 (-0.25)
	Text-only	1488 (-0.02)	1471 (-0.15)	1485 (0.02)	1492 (-0.02)
	NeuroCBT	1548 (0.96)	1563 (1.04)	1545 (0.93)	1556 (0.99)

TABLE IV
EFFECTIVENESS COMPARISON OF NEUROCBT-LLM

Model	Response Time (sec)	Analytical	Empathy	Guidance	Interactivity	Efficiency
GPT 4o	8.2 ± 3.4	1483	1456	1471	1489	179.8
GPT 4o-NeuroCBT	15.2 ± 2.1	1641	1645	1628	1635	107.7
NeuroCBT LLM	2.1 ± 0.3	1574	1589	1571	1582	751.9

TABLE V
COMPARISON OF EEG UNDERSTANDING IN NEUROCBT-LLMS

Model	Analytical	Empathy	Guidance	Interactivity	Avg
EEG	1574	1589	1571	1582	1579
Random	1456	1404	1420	1471	1438

TABLE VI
EXPERT EVALUATION RESULTS WITH STATISTICAL ANALYSIS (N = 5 EXPERTS, 100 DIALOGUE COMPARISONS)

Evaluation Criteria	Human Therapist	Our Framework
Clinical Appropriateness	4.73 ± 0.21	4.28 ± 0.34
Empathy Accuracy	4.65 ± 0.28	4.12 ± 0.41
Therapeutic Alignment	4.81 ± 0.19	4.35 ± 0.38
Overall Quality	4.69 ± 0.25	4.21 ± 0.42
Average Score	4.72 ± 0.18	4.24 ± 0.31

D. Expert Evaluation

The expert evaluation indicated that while NeuroCBT achieved clinically acceptable performance (4.24/5.0), it remained inferior to human therapists (4.72/5.0), with the most pronounced gaps observed in *Empathy* (4.12 vs. 4.65) and *Therapeutic Alignment* (4.35 vs. 4.81). Qualitative feedback highlighted several strengths, including its ability to maintain consistent therapeutic boundaries—better than some junior

clinicians, the added value of EEG integration providing objective validation for emotional assessments often based on intuition, and its methodologically sound, evidence-based CBT structure. Nonetheless, experts identified critical areas for refinement, such as improving flexibility (“sometimes applies techniques too rigidly”), enhancing cultural sensitivity in empathic responses (“misses subtle emotional nuances”), and strengthening safety mechanisms (“would benefit from more sophisticated crisis detection and escalation protocols”).

VII. CONCLUSION

This paper introduced NeuroCBT, a neuroadaptive conversational AI framework that integrates real-time EEG signals with a multi-agent CBT system grounded in Cognitive Triangle theory. By combining role-specialized agents for thoughts, feelings, and behaviors with an EEG Analysis Agent, the framework delivers emotionally aligned and context-sensitive therapeutic feedback. Key contributions include: (1) the first online CBT framework unifying multi-agent LLMs with EEG-based neurofeedback, (2) NeuroCBT-LLM, an efficient end-to-end EEG-text inference model, (3) a clinically validated dataset of 6,000 multimodal samples, and (4) enhanced interpretability via explicit reasoning chains for clinical oversight. Experiments show substantial gains in empathy, efficiency, and user satisfaction, validated by ablation studies and expert reviews. By addressing transparency and trust through interpretable architecture, NeuroCBT advances scalable, personalized, and clinically responsible digital mental health inter-

ventions, with future work exploring multi-modal integration, long-term efficacy, and advanced explainability techniques.

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